EX. NO: 8 IMPLEMENTATION OF MD5 HASHING ALGORITHM

AIM

To implement the MD5 hashing algorithm using Python.

DESCRIPTION

MD5 (Message Digest Algorithm 5) is a widely used cryptographic hash function that produces a **128-bit hash value**. It is mainly used for data integrity verification by generating a unique message digest for a given input. The same input always produces the same hash value, while even a small change in the input results in a completely different hash.

ALGORITHM

STEP-1: Import the hashlib module.

STEP-2: Read the input string from the user.

STEP-3: Convert the string into bytes.

STEP-4: Generate the MD5 hash using the md5() function.

STEP-5: Display the hexadecimal digest.

CODE

```
import hashlib


# Read input
text = input("Enter the text: ")


# Generate MD5 hash
md5_hash = hashlib.md5(text.encode())


# Display hash value
print("\nMD5 Hash Value:")
print(md5_hash.hexdigest())
```

Output

```
Enter the text: run

MD5 Hash Value:
a53108f7543b75adbb34afc035d4cdf6


=== Code Execution Successful ===
```

RESULT

Thus, the MD5 hashing algorithm was implemented successfully using Python.